



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to PSA & CO.

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

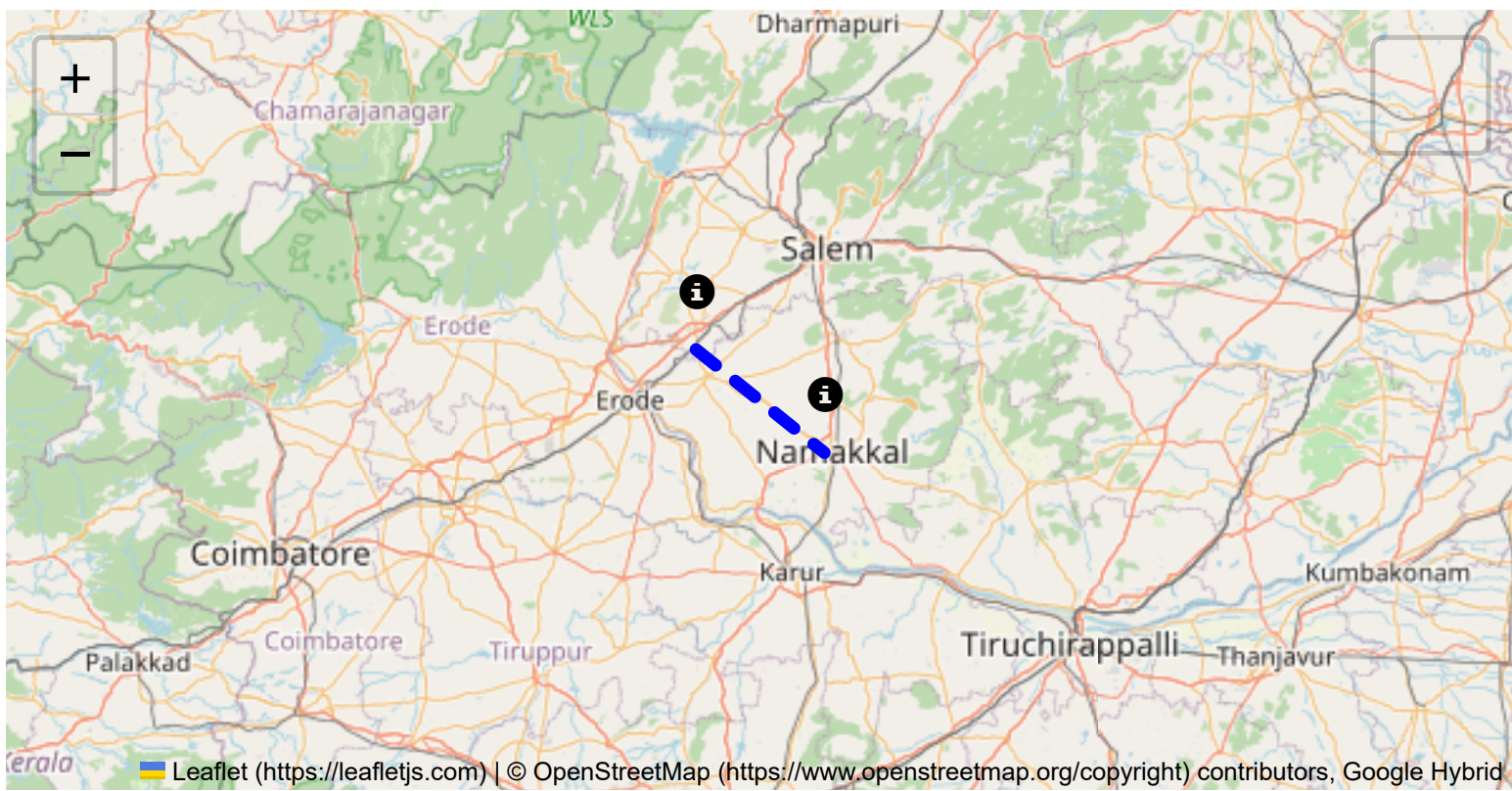
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

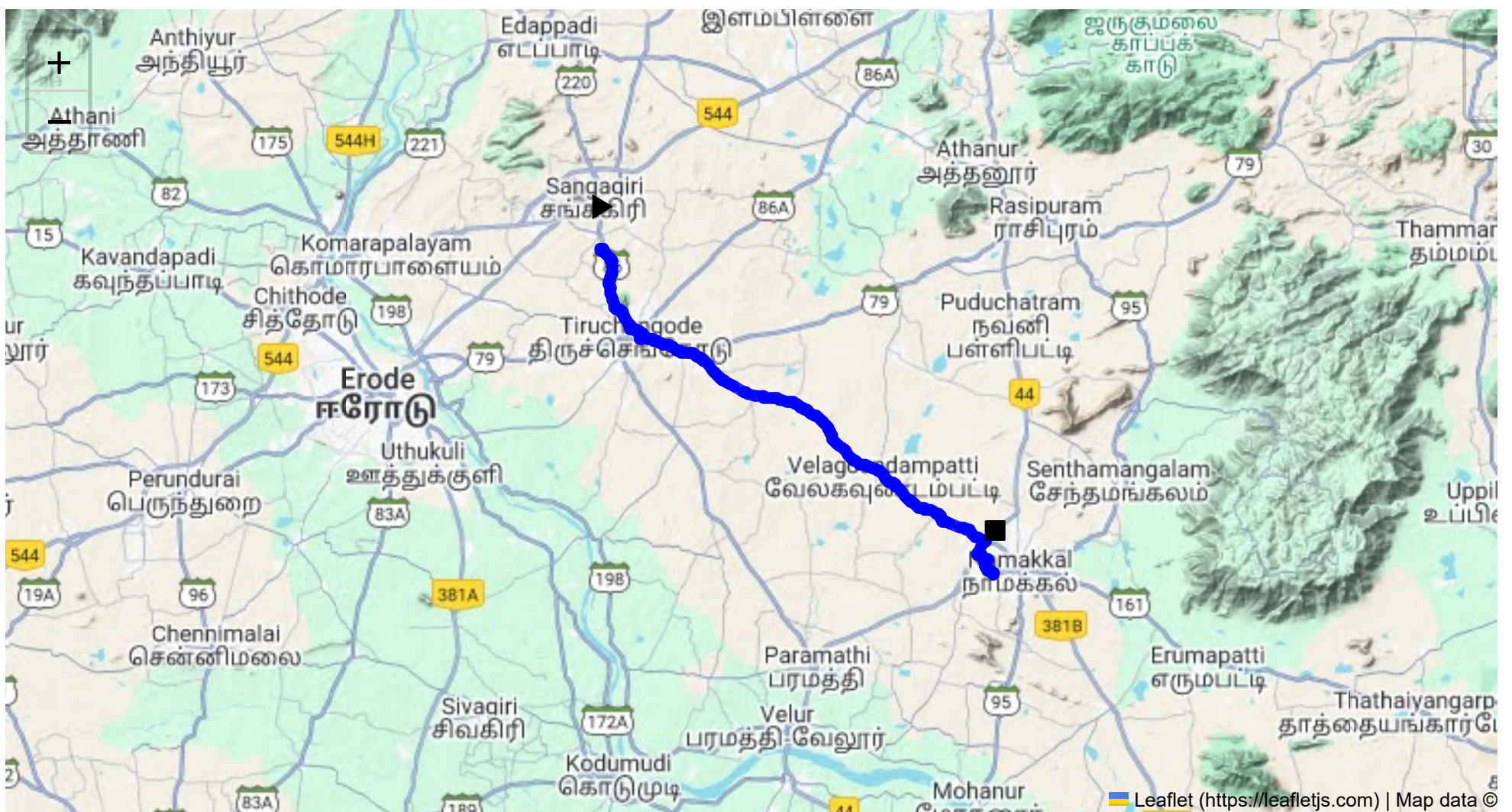
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 44.28 km
Estimated Duration: 1.0 hours
Adjusted Duration (Heavy Vehicle): 1.3 hours
Start: (11.4381, 77.8734)
End: (11.216079, 78.149048)



Welcome to the Journey Risk Management Study

Route Map Overview

The route from Sangagiri to Thillaipuram covers approximately 44.28 kilometers, taking about 1.01 hours for heavy vehicles. The path primarily involves traveling on regional highways and rural roads, passing through different small towns and rural areas.

Weather Conditions and Potential Hazards

Tamil Nadu generally experiences a tropical climate with specific challenges according to the season:

- **Monsoon (June to September):** Heavy rainfall can cause flooding and slippery roads.
- **Summer (March to June):** High temperatures can lead to road surface melting in extreme cases.
- **Winter (November to February):** Fog is less common, but there can be misty mornings that reduce visibility.

Traffic Patterns

- **Peak Hours:** Congestion can occur during morning (8:00-10:00 AM) and evening (5:00-7:00 PM) when local commuters use the roads.
- **Congestion-Prone Areas:** Smaller towns along the route, where roads narrow and traffic includes slow-moving vehicles like bicycles and tractors.

Road Quality and Infrastructure

- **Road Conditions:** Varying, with some well-maintained sections but potential potholes and uneven surfaces in rural stretches.
- **Infrastructure:** Limited roadside amenities; petrol stations and repair shops are sparse in rural sections.

Suggestions for Alternative Routes

- During extreme weather or if accidents block the main path, detours through nearby smaller rural roads might be necessary, though they may not support heavy vehicles effectively.

Local Regulations on Hazardous Material Transport

- Regulations require specific permits for hazardous materials.
- Prohibitions on transporting such materials through residential areas—careful adherence is needed when approaching towns.

Historical Incidents

- **Accidents:** There have been periodic incidents involving trucks, often due to speeding or poor road conditions.
- **Hazardous Material Incidents:** Limited data, but incidents often related to improper handling of cargo.

Environmental Considerations

- **Sensitive Areas:** Agricultural lands dominate this route; special caution is needed to avoid spills.
- **Wildlife:** Minimal wildlife crossings, but stray livestock can be a hazard.

Communication Coverage

- Coverage Along the Route:** Generally good in towns; however, some rural stretches may experience weak signals or dead zones.

Emergency Response Times

- Faster response in towns with local police and medical facilities; delayed in rural areas due to distance from services.

Overall Summary of Risk Assessment

The route presents moderate risks typical for rural Indian roads, including weather impacts, road quality issues, and variable traffic patterns. Hazardous materials require particular attention to regulations and potential environmental impacts. Communication is generally reliable but should be verified in rural segments. Overall, awareness and preparation for varying road conditions and adherence to local laws are crucial for a safe transit.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	Medium	11.43956, 77.87340	30 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr
4	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
5	Turn	Medium	11.38342, 77.89975	30 KM/Hr
6	Turn	Medium	11.38190, 77.90254	30 KM/Hr
7	Turn	High	11.38186, 77.90258	15 KM/Hr
8	Blind Spot	Blind Spot	11.37841, 77.90140	10 KM/Hr
9	Blind Spot	Blind Spot	11.23936, 78.14267	10 KM/Hr
10	Blind Spot	Blind Spot	11.23027, 78.13705	10 KM/Hr
11	Blind Spot	Blind Spot	11.22690, 78.14433	10 KM/Hr
12	Turn	Medium	11.22167, 78.14296	30 KM/Hr
13	Turn	Medium	11.22100, 78.14377	30 KM/Hr
14	Turn	Medium	11.21968, 78.14421	30 KM/Hr
15	Turn	Medium	11.21923, 78.14611	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
16	Turn	Medium	11.21861, 78.14640	30 KM/Hr
17	Turn	Medium	11.21733, 78.14809	30 KM/Hr
18	Turn	High	11.21708, 78.14889	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	Jayaa Hospital	11.3848223, 77.9021886	30 km/h	Medium
8	hospital	Vivekananda Medical Care Hospital	11.364562, 77.9482668	30 km/h	Medium
10	hospital	Manickampalayam, Government Hospital	11.32887, 78.017713	30 km/h	Medium
16	clinic	Pranauv Clinic	11.215361, 78.1484232	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
6	college	Vivekanandha Educational Institutions for Women	11.3596633, 77.9435556	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
7	college	Vivekanandha Dental College	11.3643705, 77.9475897	30 km/h	Medium
9	college	Vivekanandha College of engineering for Women	11.3565938, 77.9465342	30 km/h	Medium
11	college	The Surabi College of Polytechnic	11.2660726, 78.0939552	30 km/h	Medium
12	college	CMS College Of Engineering	11.2647508, 78.0969044	30 km/h	Medium
13	school	Porasapalayam Government School	11.2504918, 78.1163024	30 km/h	Medium
14	school	Sri Vinayaga Higher Secondary School	11.2479837, 78.1182908	30 km/h	Medium
15	school	Green Park Matric Higher Secondary School	11.2279968, 78.1480164	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.38342, 77.89975



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.38190, 77.90254



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38186, 77.90258



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.37841, 77.90140



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.23936, 78.14267



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.23027, 78.13705



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.22690, 78.14433



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.22167, 78.14296



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.22100, 78.14377



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.21968, 78.14421



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.21923, 78.14611



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.21861, 78.14640



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.21733, 78.14809
